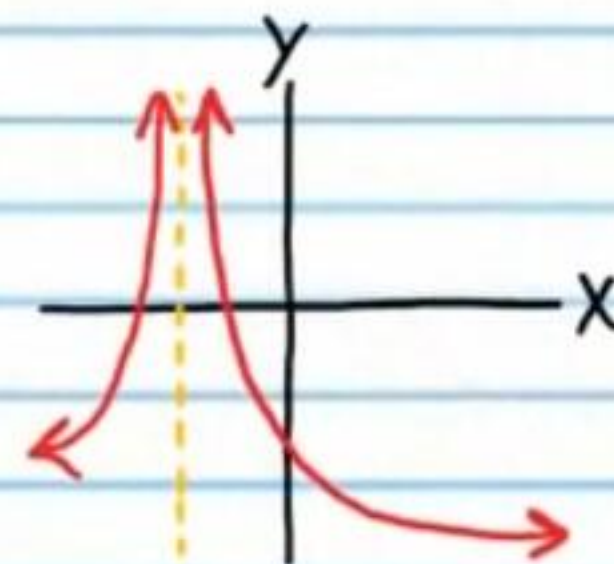
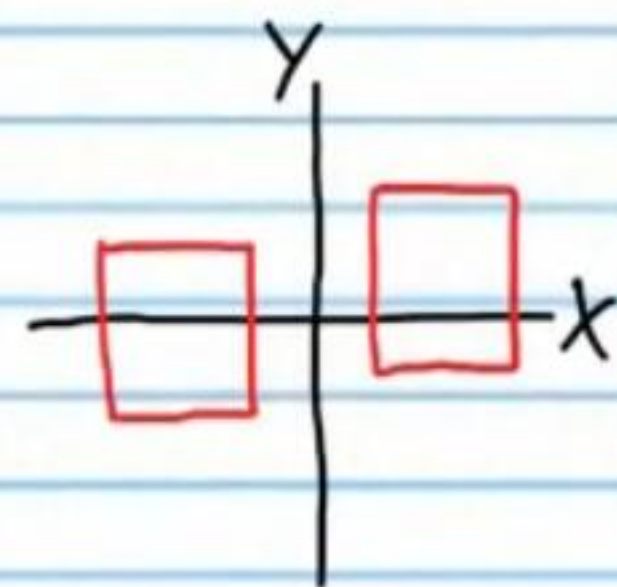


Practice Intermediate Algebra Final Test (Version 2)

- I) This is a cumulative exam.
- II) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- III) Show all work.
- IV) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- V) You will need a calculator for this test.
- VI) You do not have to write the Roman numeral for each exponent rule that you use on this test.
- VII) Do not leave your answers with fraction exponents, negative exponents, zero power exponents or 1st power exponents.
- VIII) Each problem is worth eight points for a total of 200 points.

① Circle the graph or table that cannot represent a function.



x	y
8	3
8	0
8	-1
8	4
8	5
8	-3

x	y
2	3
4	3
10	3
12	3
15	3
20	3

② i) Find the pH of a solution with the following hydrogen ion concentration. Round to the nearest tenth.

$$[H^+] = 0.002$$

ii) Simplify.

$$\sqrt[3]{\frac{1}{64^{-2}}}$$

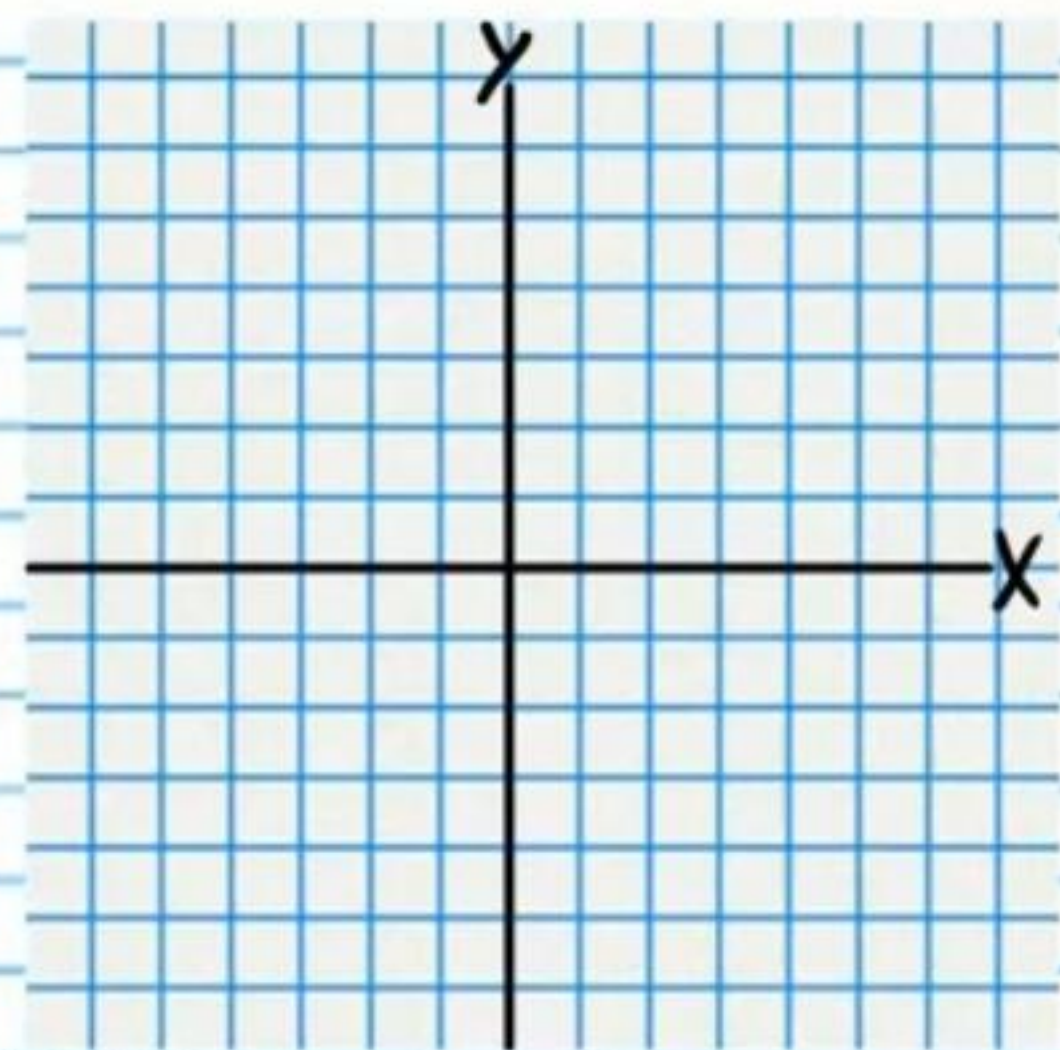
③ Write in $a+bi$ form.

$$\frac{i+2}{5i-3}$$

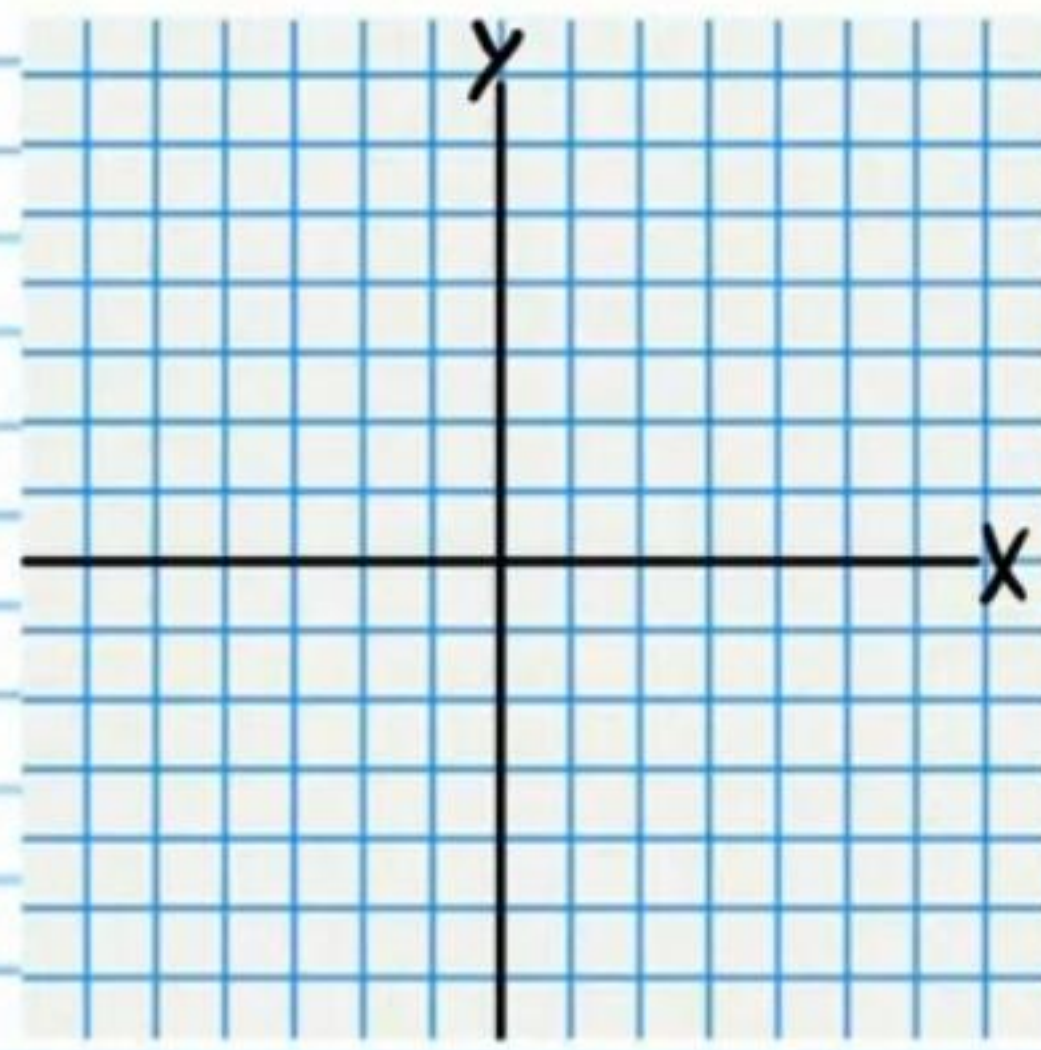
④ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation.

$$f(x) = \frac{x^3}{2}$$

Parent Function



Transformation



ii) Write the domain and range of the function in set notation.

Domain:

Range:

⑤ Solve.

$$1 = -2 \left| \frac{x}{2} - 1 \right| + 3$$

⑥ Simplify.

$$\frac{(b^3 y^2)^3 \cdot b^1 x^{-6}}{(b^2 y^6)^2 c^{-2}}$$

⑦ Solve.

$$\frac{1}{\sqrt[3]{16}} = 4^x \cdot 16$$

⑧ Solve.

$$\frac{9^x}{3^x} = 3^{2x^2-1}$$

⑨ Solve.

$$\log_7(X-4) - \log_7(X+3) = 3\log_7 2$$

⑩ Solve. Round to the nearest ten thousandth.

$$2\ln(4) + \ln(X+6) = 3$$

⑪ If $f(x) = \frac{1}{3^2-x}$ and $g(x) = \sqrt{x-3}$, Find the following:

i) $(f-g)(x)$

ii) $(g \circ f)(4)$

Find the values below if $h(x) = \begin{cases} 2^{x+1}, & -2 \leq x < 12 \\ 5, & 17 \leq x < 25 \end{cases}$

iii) $h(-1)$

iv) $h(18)$

v) $h(25)$

vi) $h(-2)$

vii) Write the domain of $h(x)$ in interval notation.

viii) Write the domain of $h(x)$ in set notation.

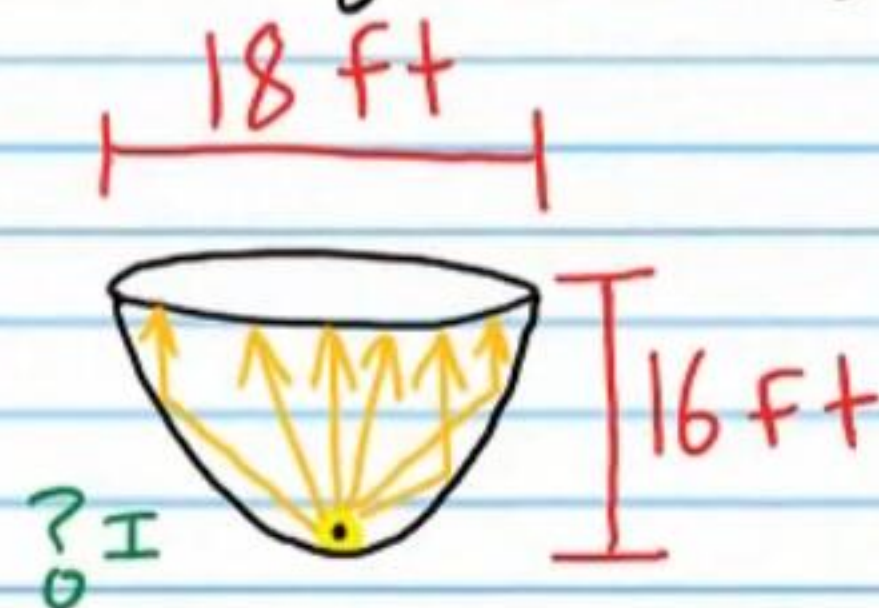
⑫ Solve by completing the square. Write in $a+bi$ form.

$$-3x^2 + 2x - 6 = 0$$

⑬ Convert to the other two forms (i.e. Standard and Vertex).

$$y = -2(x-2)(x-10)$$

(14) A searchlight is shaped like a paraboloid. If the opening is 18 feet wide and the depth of the recess is 16 feet, where should the light be placed so that the rays are reflected in the right direction?



(15) Solve. Round to the nearest thousandth.

$$e^{2x} - 5e^x + 1 = 0$$

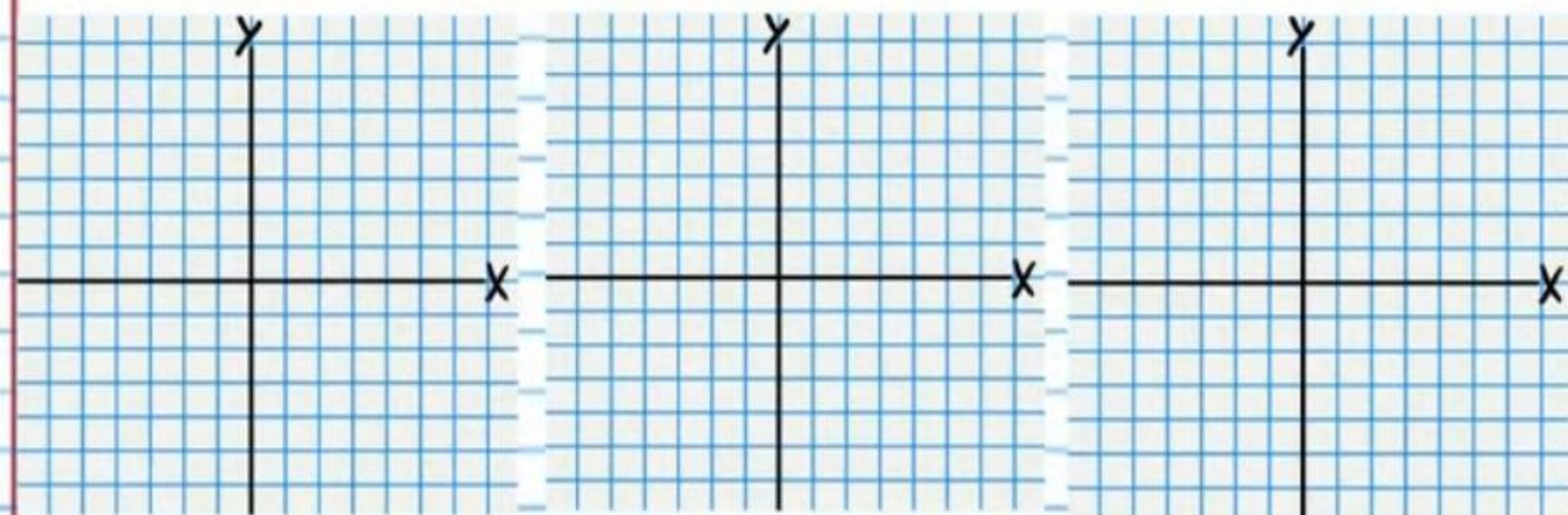
⑩ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation. Draw the asymptote(s) as dotted lines.

$$f(x) = \log_2(4x-3) + 2$$

Parent Function

Transformation #1

Final Transformation



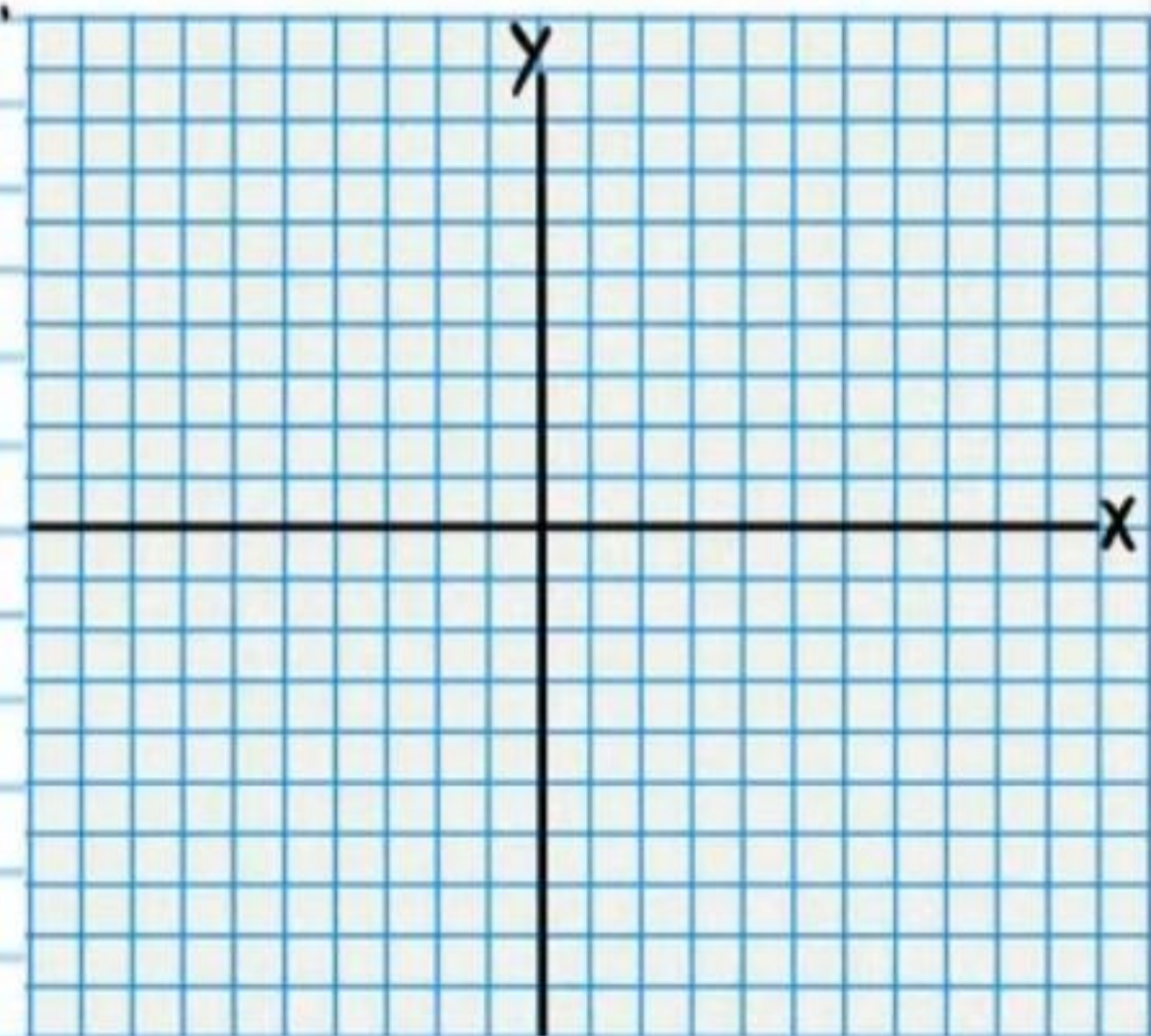
ii) Write the domain and range of the function in interval notation.

Domain:

Range:

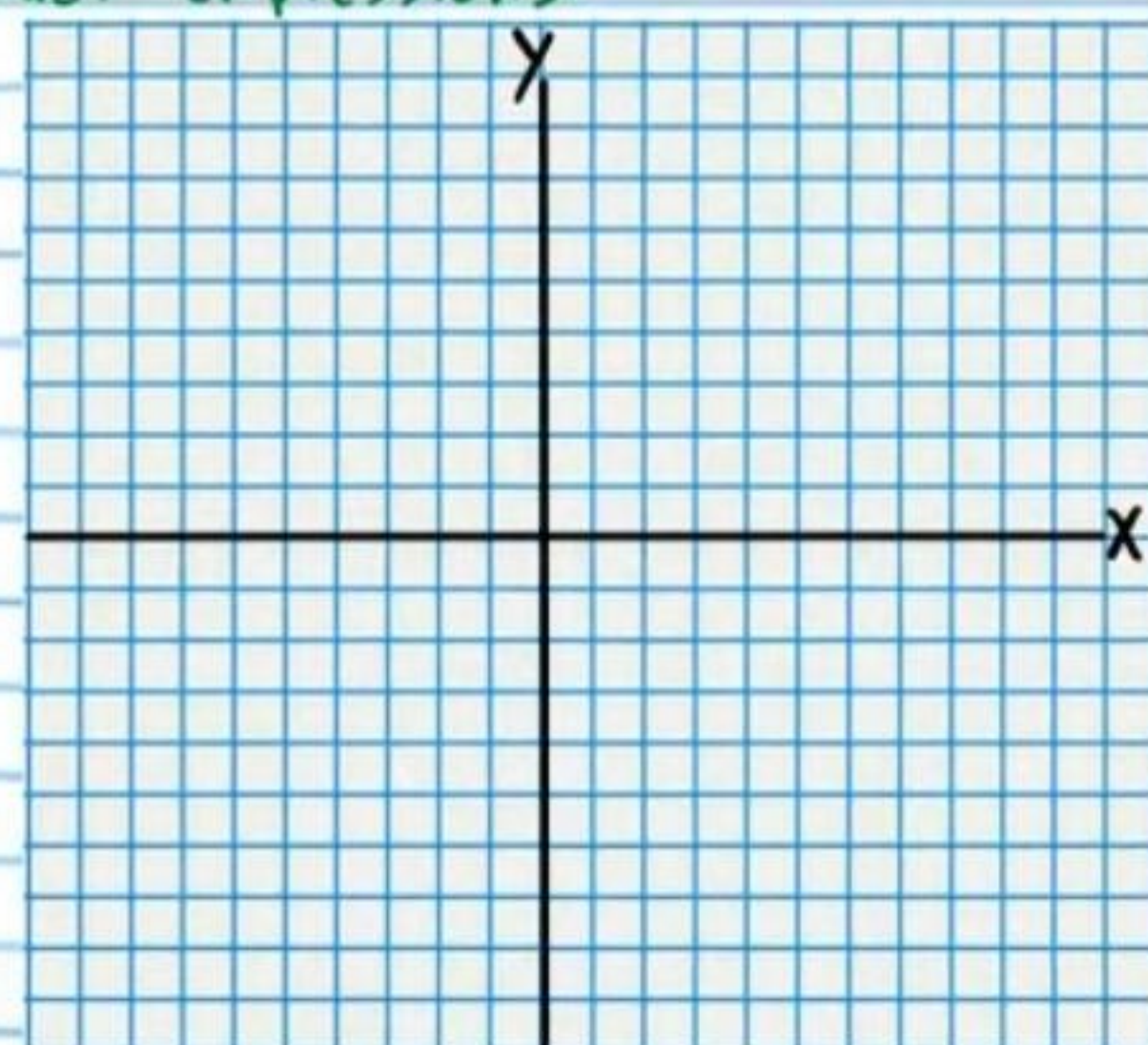
⑪ Graph the following quadratic expression. Draw the line of symmetry and write its equation. Write the coordinates of the vertex, x-intercepts, and y-intercept, and draw the points on the graph.

$$y = (x-2)(x-4)$$



⑮ Graph the following quadratic expression. Write the coordinates of the vertex, the x-intercepts, and the focus, and draw the points on the graph. Write the x-intercepts as exact expressions.

$$-2(x+3)^2 + 4 = y$$



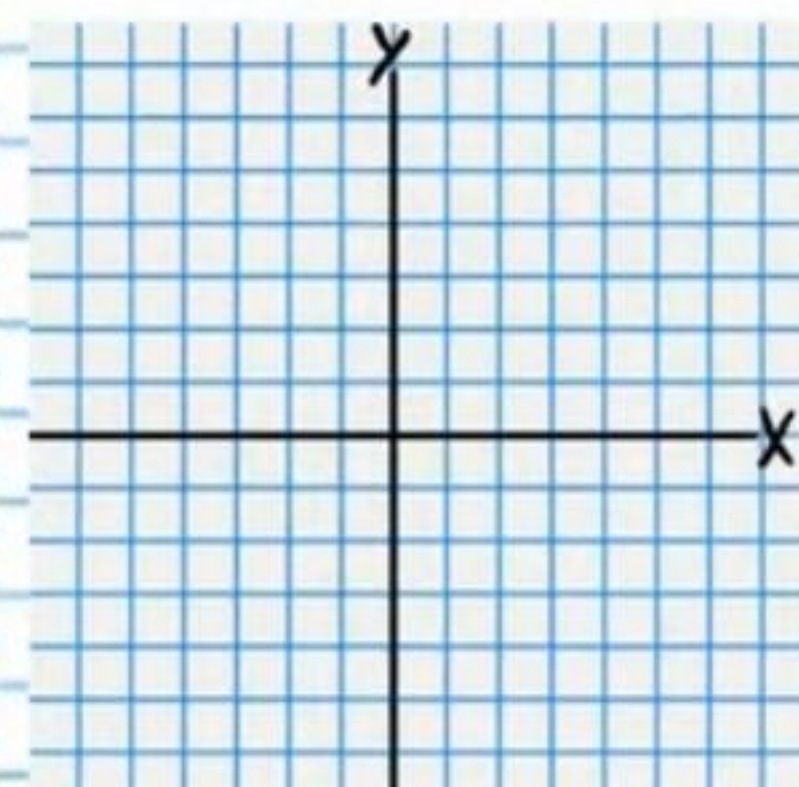
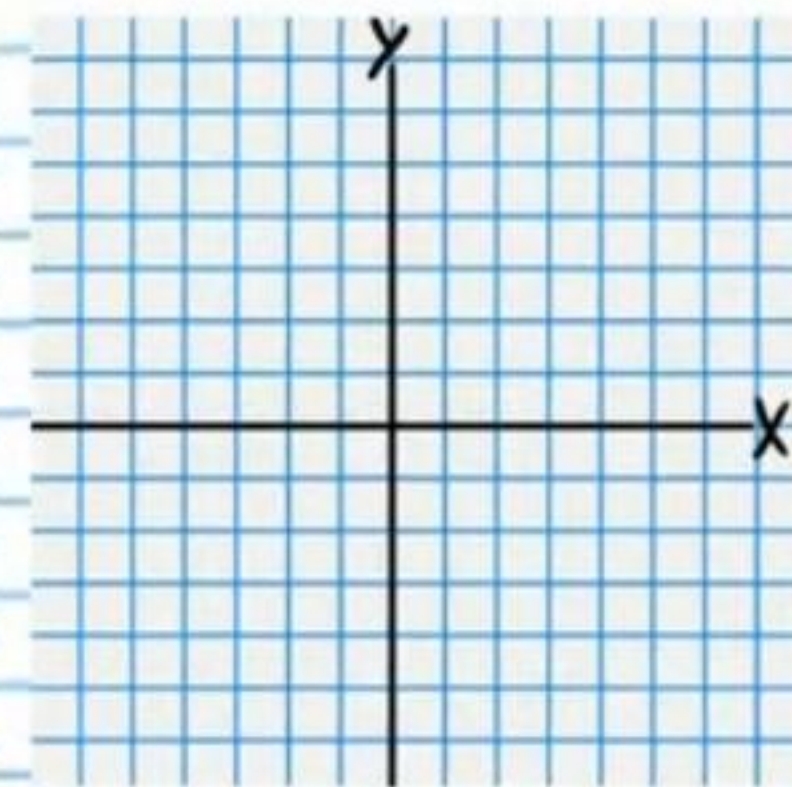
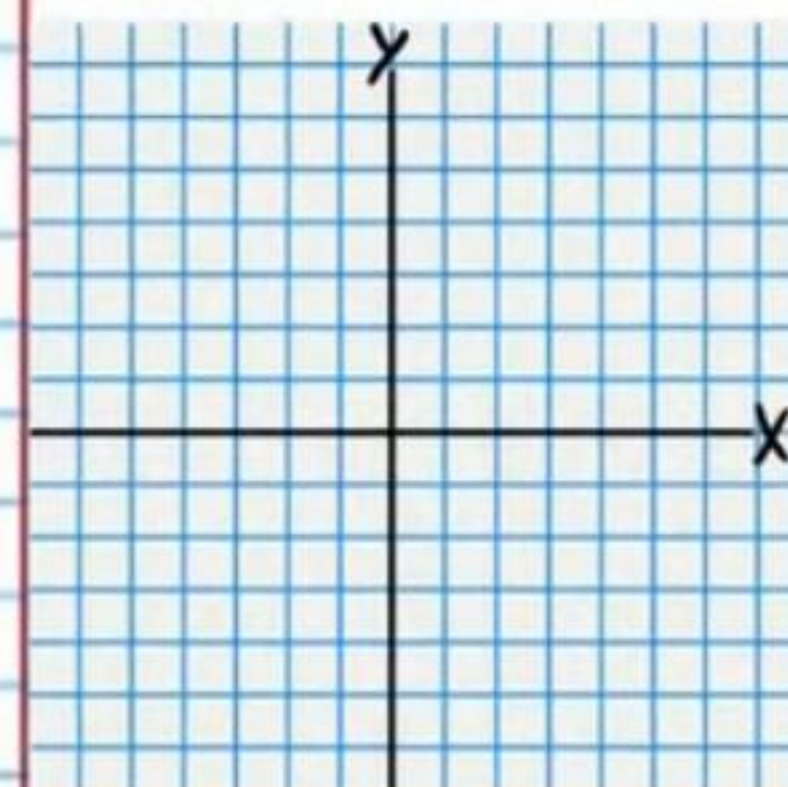
⑯ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation. Draw the asymptote(s) as dotted lines.

$$f(x) = \frac{1}{-x-1} + 5$$

Parent Function

Transformation #1

Final Transformation



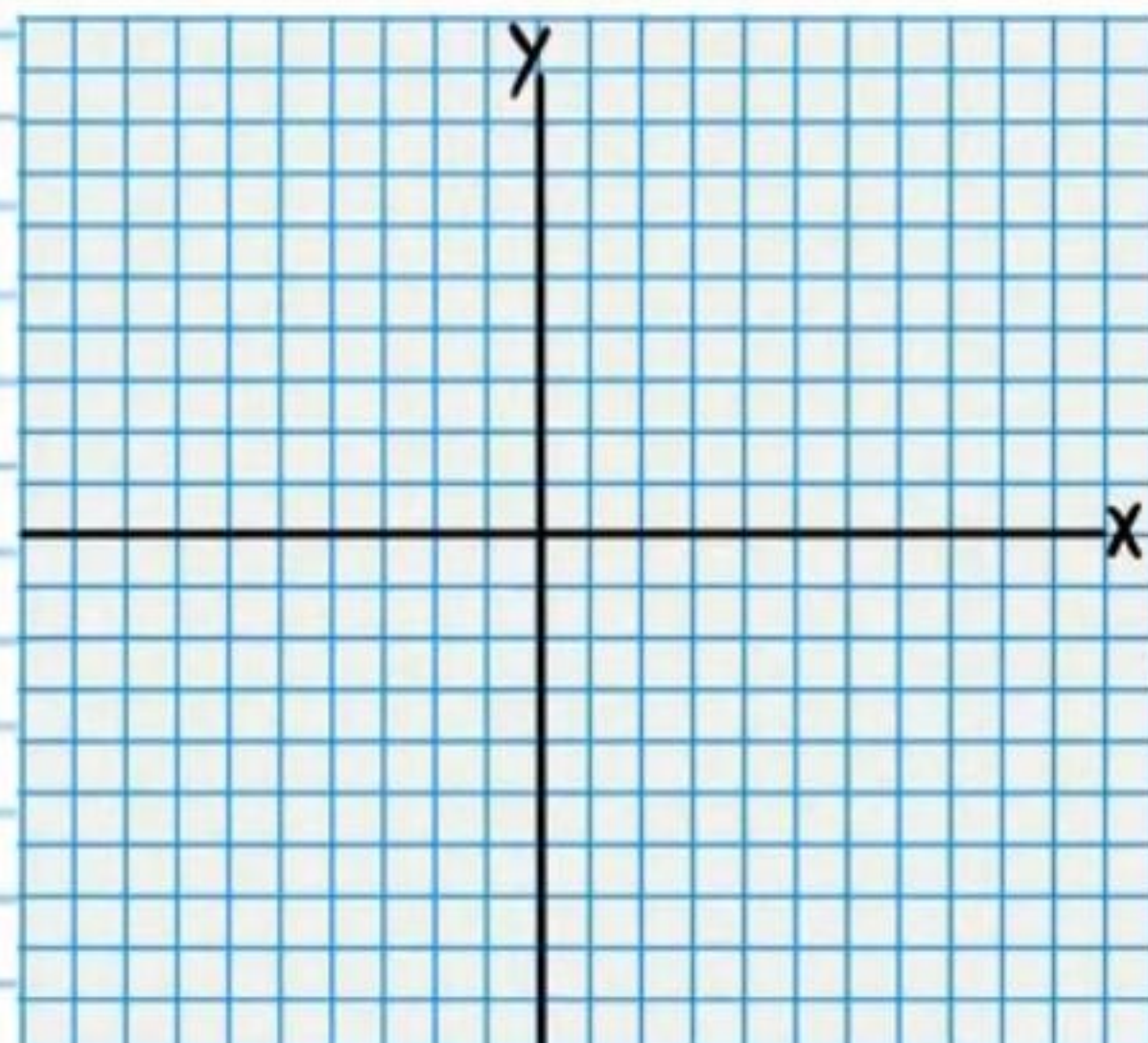
ii) Write the domain and range of the function in set notation.

Domain:

Range:

20) Graph the following quadratic expression. Write the equation of the directrix, the coordinates of the Vertex, and the coordinates of the x-intercepts (Write the x-intercepts as exact expressions). Draw the directrix as a dotted line and show each point on the graph.

$$y = \frac{1}{4}x^2 - 2x + 2$$



21) Rationalize and simplify.

$$\sqrt{\frac{25}{54x^3}}$$

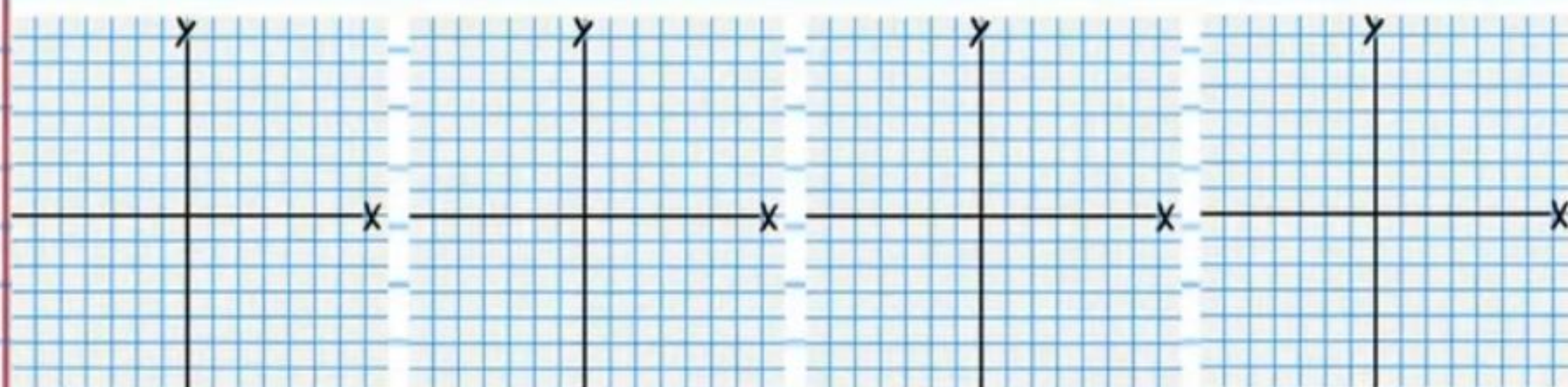
22) Simplify.

$$-3\sqrt[4]{48x^9} + 2x\sqrt[4]{243x^5}$$

Q3 i) Graph the function below by performing multiple transformations on the graph of the parent function. *Graph each transformation and write its equation.*

$$f(x) = -\frac{|x-2|}{3} - 1$$

Parent Function Transformation #1 Transformation #2 Final Transformation



ii) Write the domain and range of the function in interval notation.

Domain:

Range:

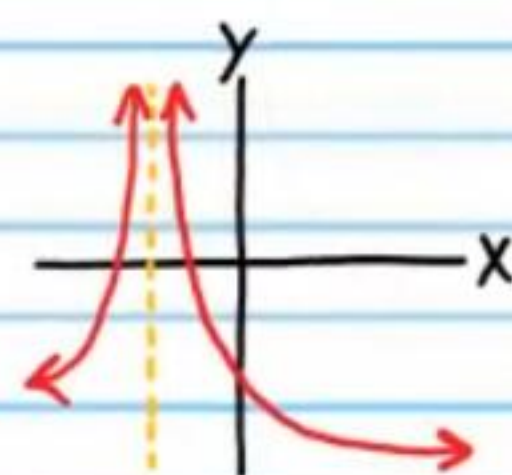
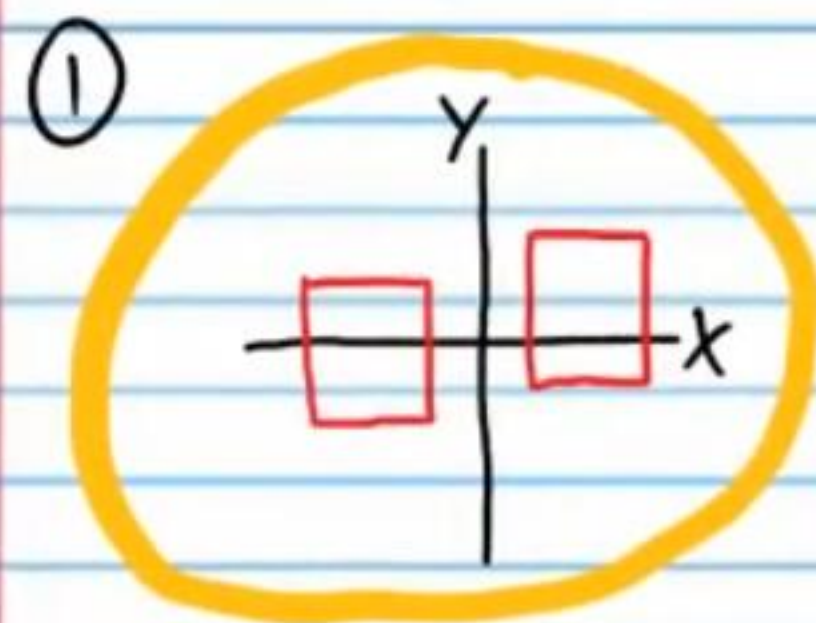
Q4 i) If \$23,000 is deposited in a bank account paying a 2.1% annual interest rate compounded monthly, how much money will be in the account in 5 years? Assume that there are no withdrawals or additional deposits.

ii) If interest is compounded continuously for 8 years at a 1% rate, how much would a \$4,000 investment grow to?

(25) Solve.

$$\begin{cases} y = x + 2z \\ x - y + z = 1 \\ -2x + y - z = -3 \end{cases}$$

Answers



② i) 2.7 ii) 16

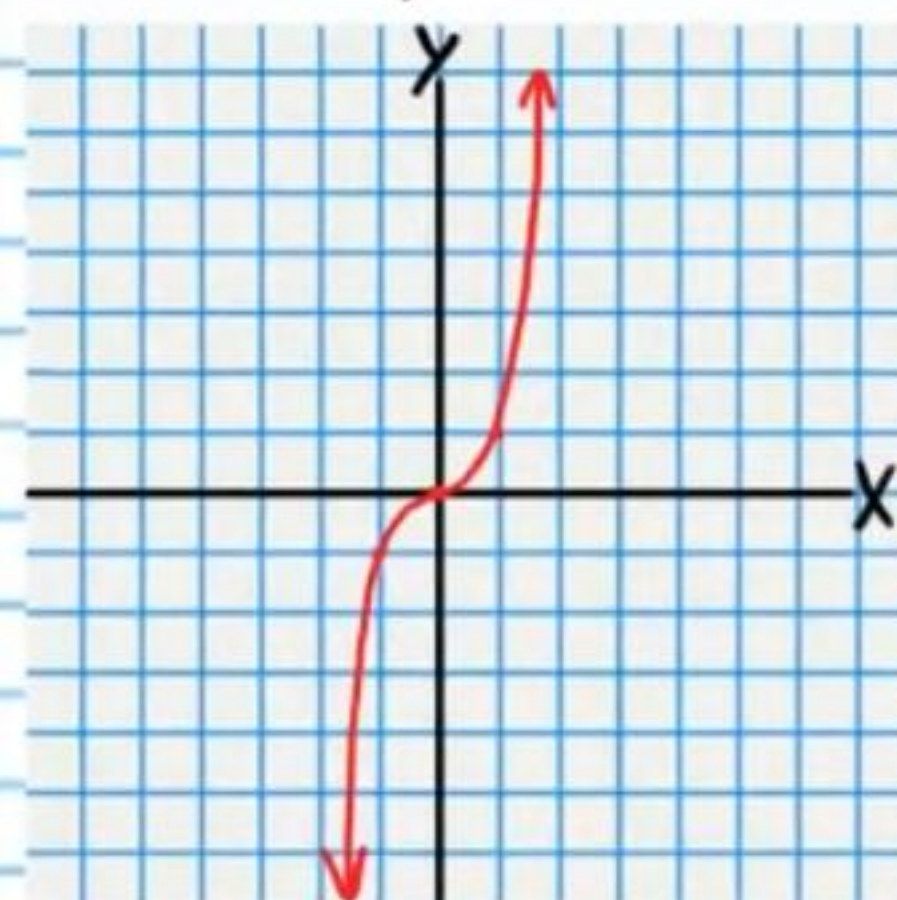
③ $-\frac{1}{34} - \frac{13}{34}i$

x	y
8	3
8	0
8	-1
8	4
8	5
8	-3

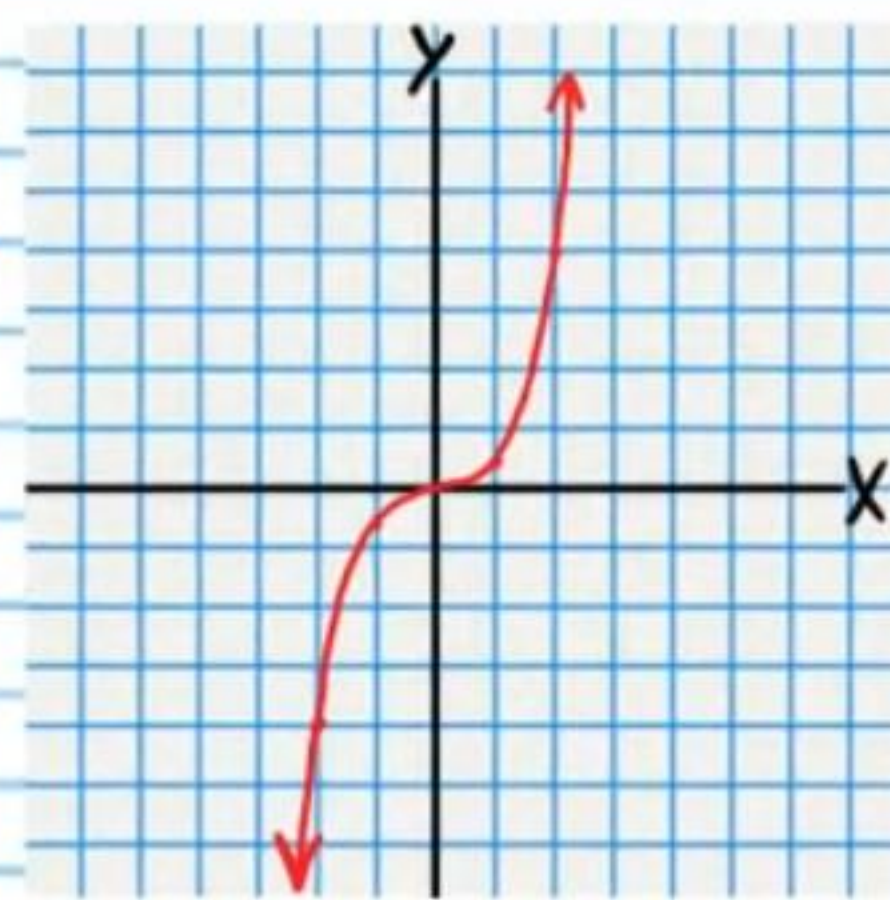
x	y
2	3
4	3
10	3
12	3
15	3
20	3

④ i)

$$y = x^3$$



$$2y = x^3$$



ii) Domain: All real numbers Range: All real numbers

⑤ $x=0$ or $x=4$

⑥ $\frac{b^6 c^2}{y^6 x^6}$ ⑦ $-\frac{8}{3} = x$ ⑧ $x = -\frac{1}{2}$ or $x=1$

⑨ $x \neq -\frac{28}{7}$ No Solution ⑩ $x \approx -4.7447$

⑪ i) $\frac{1}{32-x} - \sqrt{x-3}$ ii) $\sqrt{6}$ iii) 1 iv) 5

v) Undefined vi) $\frac{1}{2}$ vii) $[-2, 12) \cup [17, 25]$

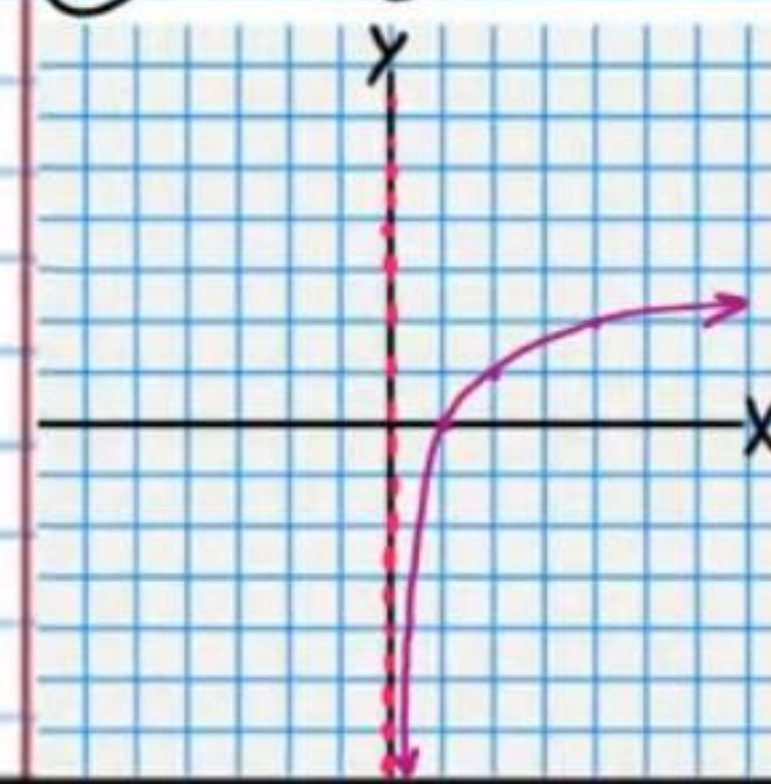
viii) $\{x \mid -2 \leq x < 12 \text{ or } 17 \leq x < 25\}$

⑫ $x = \frac{1}{3} \pm \frac{\sqrt{17}}{3}i$

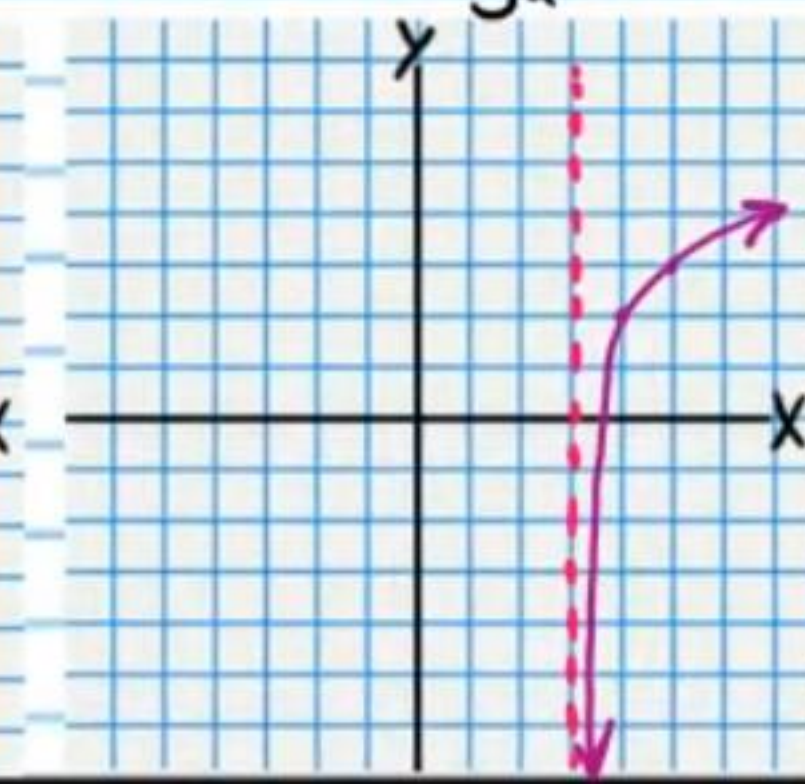
⑬ $y = -2(x-6)^2 + 32$ ⑭ $\frac{81}{64}$ ft or $1\frac{17}{64}$ ft above the vertex.

$y = -2x^2 + 24x - 40$ ⑮ $x \approx 1.567$ $x \approx -1.567$

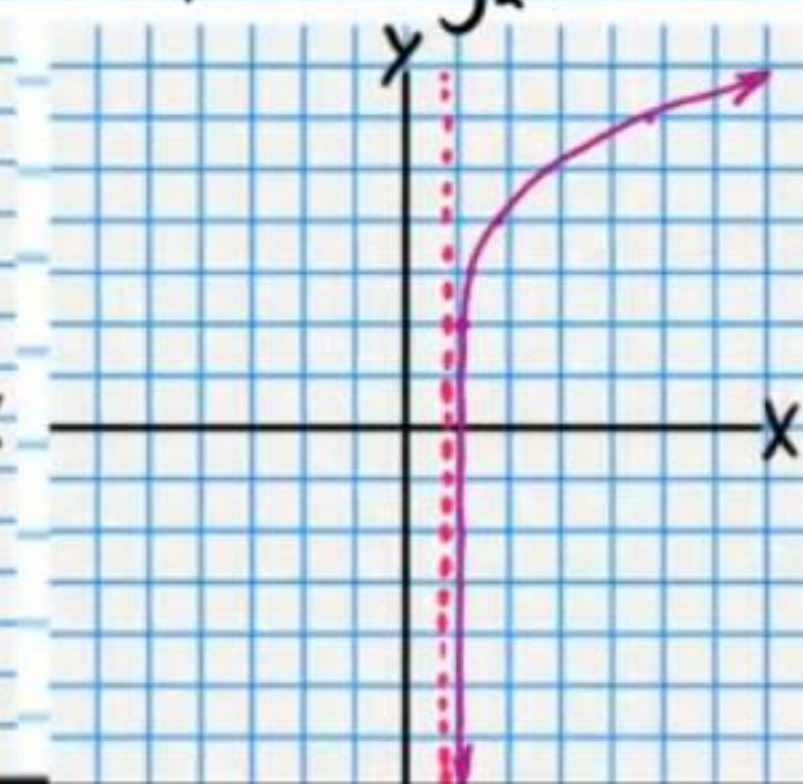
⑯ $y = \log_2 x$



$$y-2 = \log_2(x-3)$$



$$y-2 = \log_2(4x-3)$$



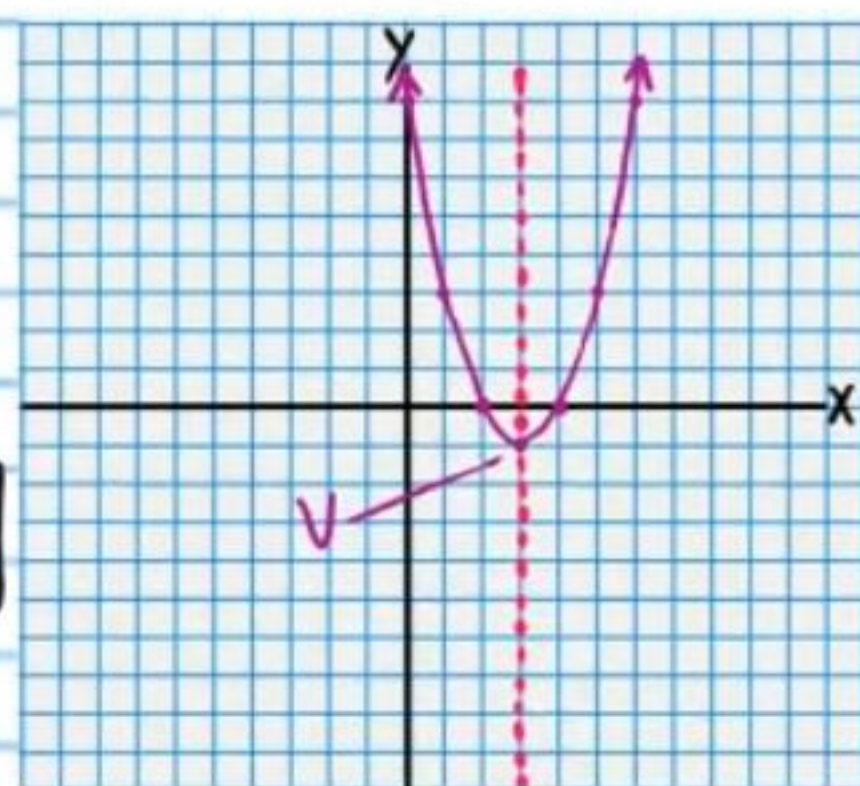
Domain: $\left(\frac{3}{4}, \infty\right)$ Range: $(-\infty, \infty)$

①⑦ $x=3$

$V(3, -1)$

$(2, 0)$ & $(4, 0)$

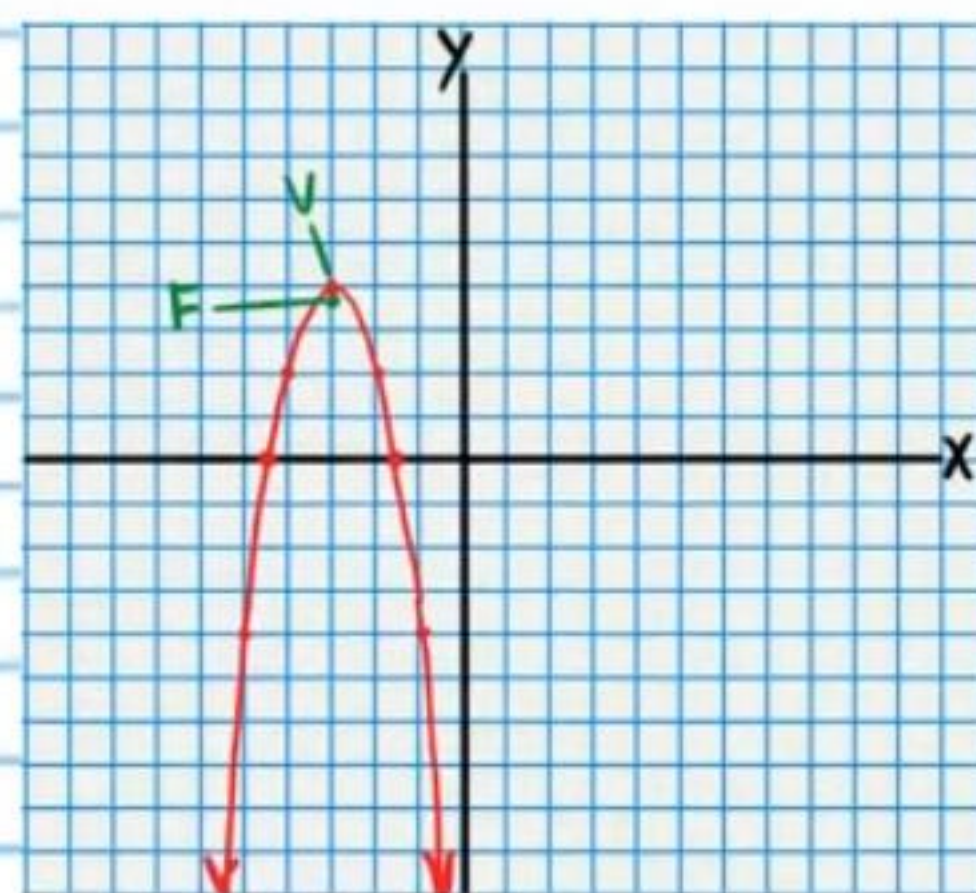
$(0, 8)$



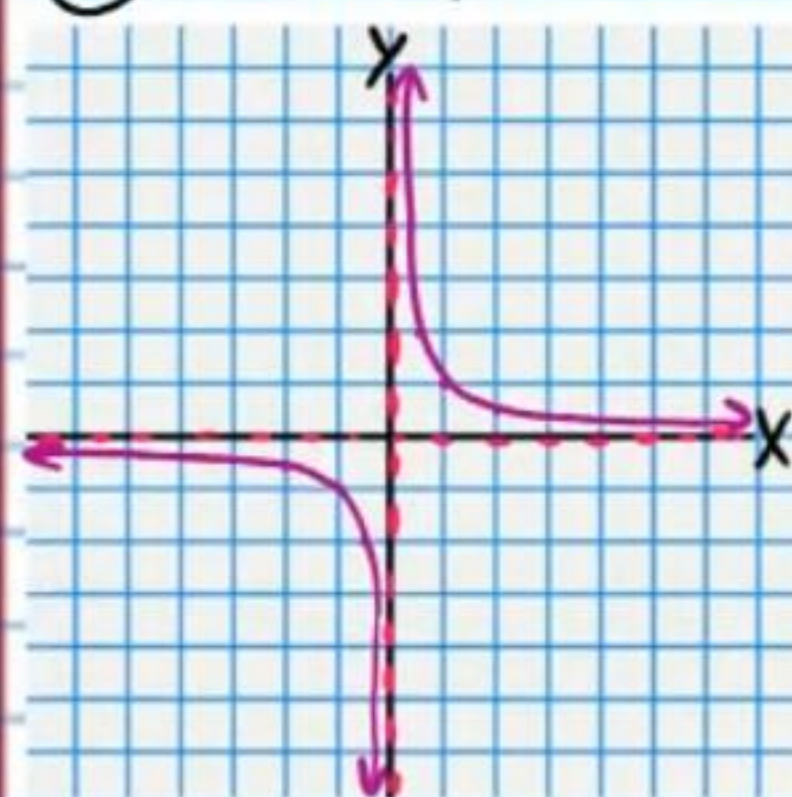
①⑧ $V(-3, 4)$

$(\sqrt{2}-3, 0)$ & $(-\sqrt{2}-3, 0)$

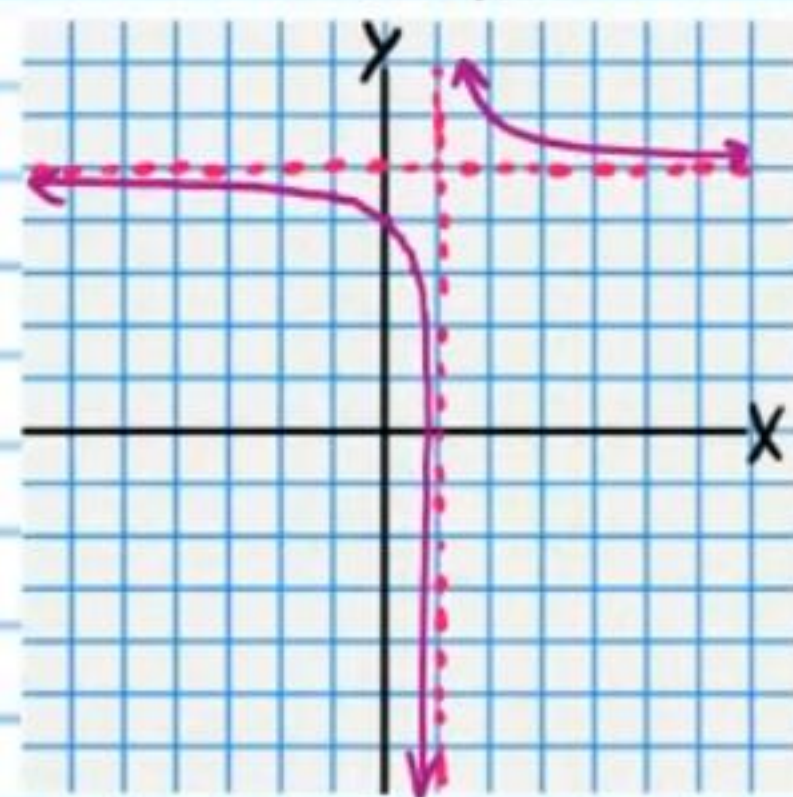
$F(-3, 3\frac{7}{8})$



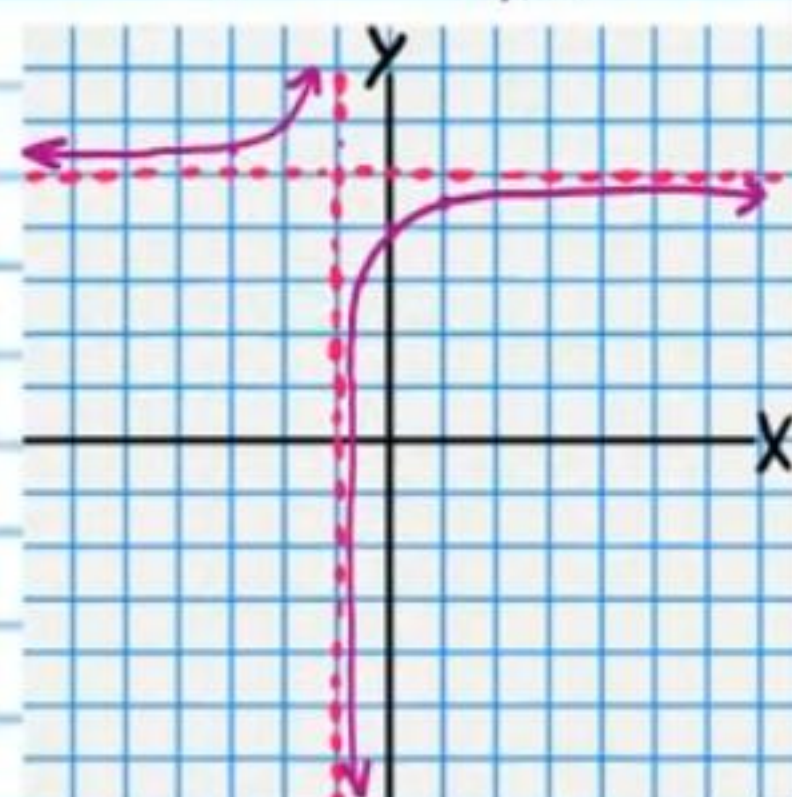
①⑨ $y = \frac{1}{x}$



$y-5 = \frac{1}{x-1}$



$y-5 = \frac{1}{-x-1}$



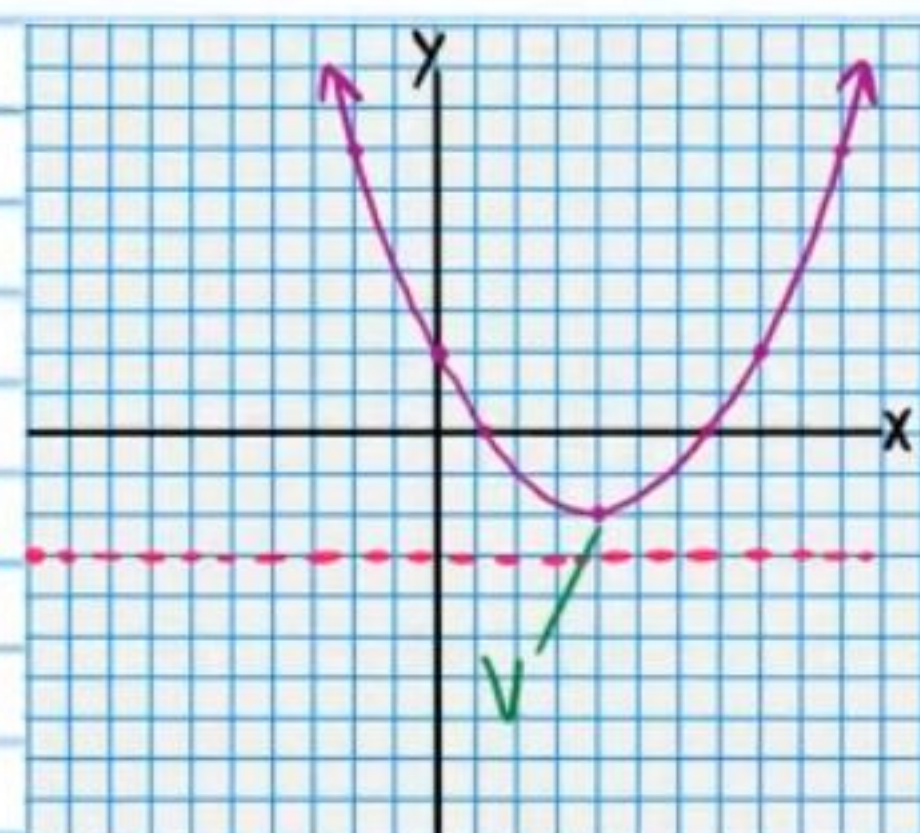
Domain: $\{x | x \neq -1\}$ Range: $\{y | y \neq 5\}$

②⑩

$y = -3$

$V(4, -2)$

$(4+2\sqrt{2}, 0)$ & $(4-2\sqrt{2}, 0)$

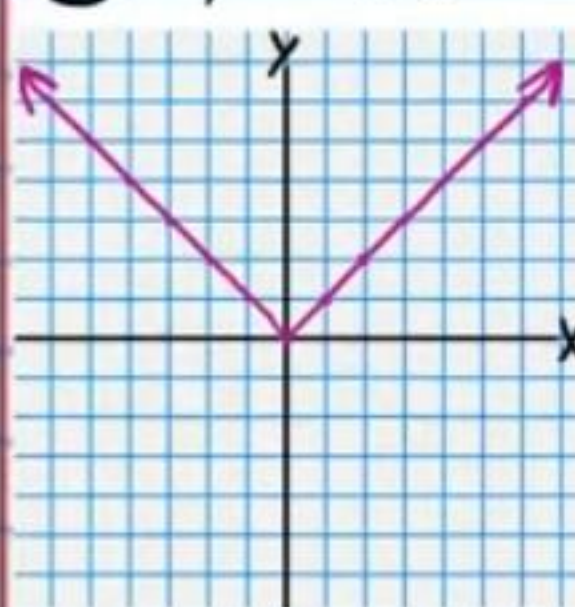


②⑪

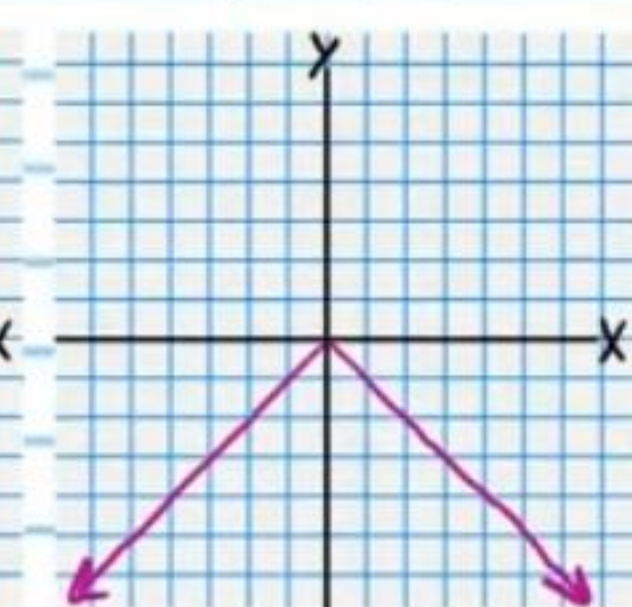
$\frac{5\sqrt{6x}}{18x^2}$

②⑫ 0

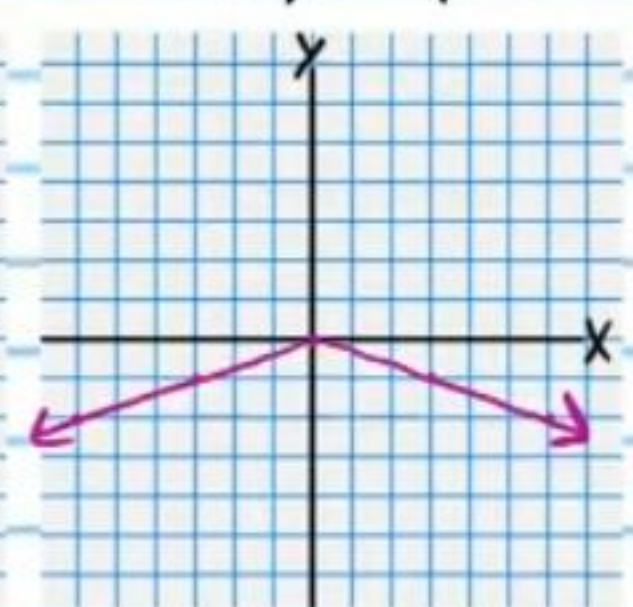
②⑬ $y = |x|$



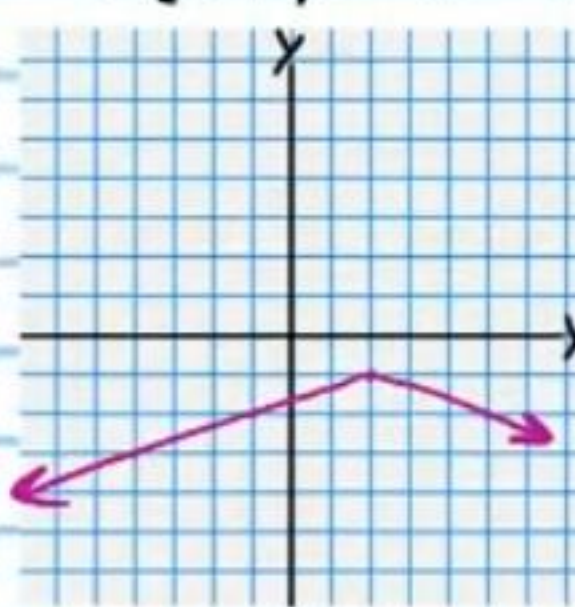
$-y = |x|$



$-3y = |x|$



$-3(y+1) = |x-2|$



Domain: $(-\infty, \infty)$ Range: $(-\infty, -1]$

②⑭ i) \$25,544

ii) \$4,333.15

②⑮ $(2, 0, -1)$